Instructor's Note:

Extra credit for the class project (Sec. 9, +10 points). The History Variable Reactive Burn (HVRB) model of G. I. Kerley is pressure-based, simple and robust, and in widespread use; it adds one progress variable and one closure rule to the hydrocode you already have. Nothing on this sheet is required, and skipping it costs you nothing.

These equations build upon U3. Continuum Mechanics & Balance Laws and the Class Project: Hydrocode sheets (Method AV and Method RS)

What HVRB Adds to Your Hydrocode:

- A progress variable, $\boldsymbol{\lambda}$, in every Material X cell: $\boldsymbol{\lambda = 0}$ unreacted, $\boldsymbol{\lambda = 1}$ fully reacted (carried with the cell; a Lagrangian mesh needs no advection)
- A rate law, pressure history $\rightarrow \boldsymbol{\lambda}$ pages 1–3
- A closure, $(\boldsymbol{\rho}, \boldsymbol{e}, \boldsymbol{\lambda}) \rightarrow \boldsymbol{P}$ page 4
- Notation, history variable $\boldsymbol{\phi}$, reaction potential $\boldsymbol{\Phi}$, topology function $\boldsymbol{g}(\boldsymbol{\lambda})$; phases $(\ )_{\text{solid}}$ and $(\ )_{\text{gas}}$; time level $\boldsymbol{n}$
- This $\boldsymbol{\lambda}$ is not the U7 wave speed $\boldsymbol{\lambda_i}$, nor the U4 Lamé constant (it is the mass fraction of reaction product/gas, more closely related to U3's $\boldsymbol{Y_k}$, $\dot{\boldsymbol{\Omega}}_{\boldsymbol{k}}$, and $\boldsymbol{Q_k}$)
- ***Everything else in your cycle is unchanged***

Rate Law — Separable, History-Variable Form:

- Separable assumption, $\dfrac{\boldsymbol{D\lambda}}{\boldsymbol{Dt}} = \boldsymbol{g}(\boldsymbol{\lambda})\ \boldsymbol{\Phi}(\boldsymbol{P})$ $\boldsymbol{g}$ depends on $\boldsymbol{\lambda}$ alone, $\boldsymbol{\Phi}$ not on $\boldsymbol{\lambda}$
- History variable, $\boldsymbol{\phi}(\boldsymbol{t}) = \displaystyle\int_0^t \boldsymbol{\Phi}\, \boldsymbol{d\hat{t}}$ so that $\boldsymbol{g}(\boldsymbol{\lambda}) = \dfrac{\boldsymbol{d\lambda}}{\boldsymbol{d\phi}}$
- Inversion, $\boldsymbol{\phi} = \boldsymbol{F}(\boldsymbol{\lambda}) = \displaystyle\int_0^{\lambda} \frac{\boldsymbol{d\hat{\lambda}}}{\boldsymbol{g}\left(\hat{\boldsymbol{\lambda}}\right)}$ and $\boldsymbol{\lambda} = \boldsymbol{F^{-1}}(\boldsymbol{\phi})$
- ***The extent of reaction depends only on the accumulated pressure history, $\boldsymbol{\phi}$, not on how the pressure was delivered***

HVRB Topology (M_R, X_R) [1, 2, 3, 4], Fig. 1:

- Extent of reaction, $\lambda = 1 - \left(1 - \dfrac{\phi^{M_R}}{X_R}\right)^{X_R}$ for $0 \le \phi \le \phi_{end} = X_R^{1/M_R}$, then $\lambda = 1$
- History variable, $\phi = \left[X_R\left(1 - (1-\lambda)^{1/X_R}\right)\right]^{1/M_R}$
- Topology function, $g(\lambda) = \dfrac{d\lambda}{d\phi}$

  . $= M_R X_R^{1-1/M_R} (1-\lambda)^{1-1/X_R}\left[1-(1-\lambda)^{1/X_R}\right]^{1-1/M_R}$
- Shape, $g(0) = 0$ needs $M_R > 1$ (a delay behind the shock); $g(1) = 0$ needs $X_R > 1$ (a gradual finish)

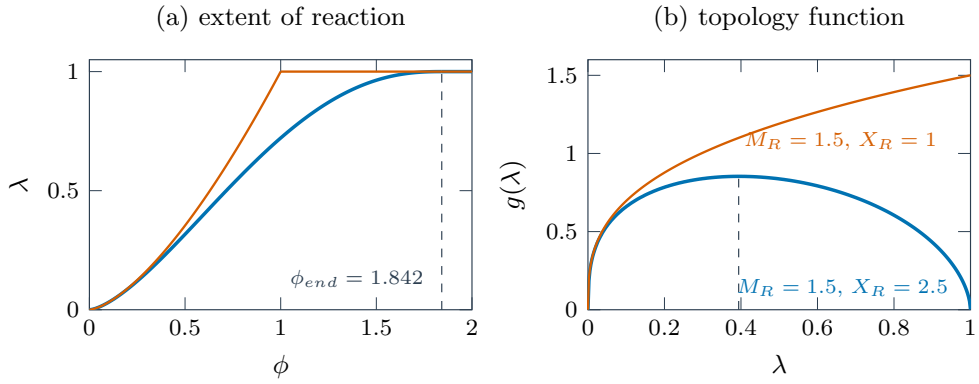


Figure 1: HVRB topology, exact curves. (a) $\lambda(\phi)$; (b) $g(\lambda) = d\lambda/d\phi$. Blue: the Material X set, $M_R = 1.5$, $X_R = 2.5$, with g peaking at $\lambda = 0.394$ ($g_{max} = 0.854$) and the burn complete at $\phi_{end} = 1.842$. Orange: $M_R = 1.5$, $X_R = 1$, common input defaults; g never returns to zero and the burn stops abruptly at $\phi = 1$.

HVRB Reaction Potential (P_R, Z_R, P_I, τ_0) [2, 3]:

- Potential, $\Phi = \frac{1}{\tau_0}\left[\frac{\max(P - P_I,\ 0)}{P_R}\right]^{Z_R}$ P is the cell's mixture pressure, P^n
- ϕ grows only while $P > P_I$ and never decreases; $\tau_0 = 1$ μs by convention (not 10^{-6} in μs units)
- Only the product $\tau_0 P_R^{Z_R}$ enters: changing τ_0 is the same as rescaling P_R by τ_0^{1/Z_R} [5]
- ***HVRB cannot quench and cannot desensitize: any cell held above P_I eventually burns*** [6]**.**

Unreacted Material X (Mie–Grüneisen, from the project):

- Pressure, $P_{\text{solid}}(\mathrm{V}, e) = \rho_0 \Gamma_0 e + \left[\frac{1}{\mathrm{V}_0 - \mathrm{V}} - \frac{1}{2}\rho_0 \Gamma_0\right] u_p^2$ project Eq. (4), $\Gamma\rho = \Gamma_0\rho_0$
- Principal isentrope, $\frac{de_S}{d\mathrm{V}} = -P_{\text{solid}}(\mathrm{V}, e_S)$, $e_S(\mathrm{V}_0) = 0$; $T_S = T_0 \exp(\Gamma_0 \eta)$
- Temperature, $e_{\text{solid}} = e_S(\mathrm{V}) + C_{\mathrm{V}}[T - T_S(\mathrm{V})]$ $e = 0$, $T = T_0$ at rest
- Here T is only an intermediate result that is needed by the closure rule

Detonation Products (constant γ, from the project):

- Pressure, $P_{\text{gas}} = (\gamma - 1)\rho(e_{\text{gas}} + Q)$ $\gamma = 2s_1 - 1 = 2.60$, $Q = 4.8828$ MJ/kg, project Eq. (8)
- Temperature, $e_{\text{gas}} = C_{\mathrm{V}} T - Q$ so $P_{\text{gas}} = (\gamma - 1)\rho C_{\mathrm{V}} T$
- Same C_{V} as the solid, for simplicity's sake in your hydrocode

Mixture Closure (density–temperature equilibrium) [7, 8]:

- Both phases at the same cell density $\boldsymbol{\rho}$ and temperature $\boldsymbol{T}$; the solid and gas phase pressures differ (pressure–temperature equilibrium is the common alternative [7])
- Pressure, $\boldsymbol{P = (1-\lambda)\, P_{\text{solid}}(\rho, T) + \lambda\, P_{\text{gas}}(\rho, T)}$
- Energy, $\boldsymbol{e = (1-\lambda)\, e_{\text{solid}}(\rho, T) + \lambda\, e_{\text{gas}}(\rho, T)}$
- One $\boldsymbol{C_{\text{V}}}$: the energy line is linear in $\boldsymbol{T}$, so given $\boldsymbol{(\rho,\ e,\ \lambda)}$ there is no iteration:
  . $\boldsymbol{C_{\text{V}} T = e - (1-\lambda)\left[e_S(\text{V}) - C_{\text{V}} T_S(\text{V})\right] + \lambda Q}$
- Limits, $\boldsymbol{\lambda = 0}$ returns $\boldsymbol{P_{\text{solid}}(\text{V}, e)}$; $\boldsymbol{\lambda = 1}$ returns $\boldsymbol{(\gamma-1)\,\rho\,(e+Q)}$
- Compressed solid is stiffer than gas at equal $\boldsymbol{\rho}$ and $\boldsymbol{T}$, so early burning can *raise* the pressure behind the shock
- Same closure algorithm without using $\boldsymbol{T}$ (exact, since both phases share one $\boldsymbol{C_{\text{V}}}$): let $\boldsymbol{\vartheta \equiv C_{\text{V}} T}$ and the solid's $\boldsymbol{T = 0}$ 'cold' curve,
  . $\boldsymbol{e_c(\text{V}) = e_S(\text{V}) - C_{\text{V}} T_S(\text{V}), \qquad P_c(\text{V}) = P_{\text{solid}}(\text{V}, e_S) - \rho_0 \Gamma_0 C_{\text{V}} T_S(\text{V})}$
  . $\boldsymbol{\vartheta = e - (1-\lambda)\, e_c(\text{V}) + \lambda Q}$
  . $\boldsymbol{P = (1-\lambda)\left[P_c(\text{V}) + \rho_0 \Gamma_0 \vartheta\right] + \lambda(\gamma-1)\,\rho\,\vartheta}$

Adding Burn to One Cycle (Lagrangian, Method AV or RS):

- After momentum, position, volume and energy, for each Material X cell:
- Potential, $\boldsymbol{\Phi^n}$ from $\boldsymbol{P^n}$; history, $\boldsymbol{\phi^{n+1} = \min\left(\phi^n + \Phi^n\, \Delta t^n,\ \phi_{end}\right)}$
- Progress, $\boldsymbol{\lambda^{n+1} = F^{-1}\left(\phi^{n+1}\right)}$ exact; $\boldsymbol{0 \le \lambda \le 1}$ automatically
- Pressure, closure with $\boldsymbol{\left(\rho^{n+1},\ e^{n+1},\ \lambda^{n+1}\right)}$; then $\boldsymbol{q}$ and $\boldsymbol{\Delta t}$ as usual
- Burning changes $\boldsymbol{\lambda}$ at fixed $\boldsymbol{\rho}$ and $\boldsymbol{e}$: the energy update is the one on the Hydrocode sheets, and total energy must stay conserved
- Time step, the mixture sound speed at fixed $\boldsymbol{\lambda}$ in the CFL condition
- Zoning, the steady reaction zone lasts $\boldsymbol{\approx 19}$ ns, $\boldsymbol{\approx 0.14}$ mm of initial coordinate: $\boldsymbol{\approx 5}$ cells at 35 zones/mm
- ***The stiff part, $\boldsymbol{g(\lambda)}$, is integrated exactly; the only error is the first-order quadrature of $\boldsymbol{\Phi}$***

Mixture Sound Speed (fixed $\boldsymbol{\lambda}$):

- Phases, $\boldsymbol{c}_{\text{solid}}$: inert Mie–Grüneisen at $(\text{V}, \boldsymbol{e_c + \vartheta})$; $\boldsymbol{c^2_{\text{gas}} = \gamma\,(\gamma - 1)\,\vartheta}$
- Mass-weighted, $\boldsymbol{c^2 \;\approx\; (1-\lambda)\,c^2_{\text{solid}} \;+\; \lambda\,c^2_{\text{gas}}}$ recommended, $\boldsymbol{\approx 1}\%$ high
- Exact, subtract $\boldsymbol{\lambda\,(1-\lambda)\,(\rho_0 \Gamma_0/\rho - \gamma + 1)^2\,\vartheta}$ (the spread of the two $\boldsymbol{\Gamma}$'s)
- Schmidt, $\boldsymbol{Z^{-2} = (1-\lambda)\,Z^{-2}_{\text{solid}} + \lambda Z^{-2}_{\text{gas}}}$, $\boldsymbol{Z = \rho c}$; equal-*pressure* rule, $\boldsymbol{\approx 15}\%$ low
- ***The sound speed sets only $\boldsymbol{\Delta t}$ and $\boldsymbol{q}$, not the converged solution***

HVRB Parameters for Material X (notional):

Parameter	Value	Units
$\boldsymbol{P_R}$, reference pressure	7.0	GPa
$\boldsymbol{Z_R}$, pressure exponent	3.0	–
$\boldsymbol{M_R}$, delay exponent	1.5	–
$\boldsymbol{X_R}$, completion exponent	2.5	–
$\boldsymbol{P_I}$, threshold pressure	1.0	GPa
$\boldsymbol{\tau_0}$, time constant	1.0	$\boldsymbol{\mu}$s
$\boldsymbol{C_V}$, both phases	1.16	J/(g K)
$\boldsymbol{T_0}$, initial temperature	298	K

Checks Before the Velocity Sweep:

- Inert limit, $\boldsymbol{\Phi \equiv 0}$ reproduces your inert PDV trace exactly
- Constant-volume burn, one cell at rest, $\boldsymbol{\lambda}$: $\boldsymbol{0 \to 1}$ at fixed $\boldsymbol{e}$, ends at $\boldsymbol{P = (\gamma - 1)\,\rho_0 Q = 12.50}$ GPa
- Energy, total energy conserved to $\boldsymbol{\lesssim 0.01}\%$ while Material X burns
- HVRB, $\boldsymbol{\lambda}$ rebuilt from one tracer's $\boldsymbol{P\,(t)}$ history with the two HVRB boxes matches your $\boldsymbol{\lambda\,(t)}$
- Detonation, the shock speed never settles below your $\boldsymbol{D_{CJ}}$ from project Sec. 7
- Pop plot, run distance versus input pressure is a straight line on log–log axes [9]
- Sweep, 2.00, 1.75, 1.50 km/s detonate (run $\boldsymbol{\approx}$ 3.5, 5.5, 7.5 mm); 1.25, 1.00 km/s do not within 10 mm (late PDV wave only)

References:

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[5] J. Starkenberg, "Shock-pressure and pseudo-entropic approaches to explosive initiation modeling," in *Proceedings of the Fifteenth International Detonation Symposium*, Office of Naval Research, ONR 43-280-15 (2014), pp. 908–916.

[6] L. W. Tuttle, R. G. Schmitt, D. E. Kittell, and E. N. Harstad, "Evaluation of XHVRB for capturing shock desensitization," *AIP Conf. Proc.* **1979** (2018), doi:10.1063/1.5044916.

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[8] D. E. Kittell, R. G. Schmitt, L. W. Tuttle, and E. N. Harstad, "Implementation of a CREST multistate reactive burn model in CTH for two solid high explosives," in *Proceedings of the Sixteenth International Detonation Symposium*, ONR-43-5762-19 (2018), pp. 1021–1031.

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